

LIMITLESS NATURALS BY EVA PHARMA

TECHNICAL & BUSINESS PROPOSAL

Cross-Platform Mobile Application (iOS & Android) & Full-Stack Web Platform

High-Availability Enterprise Solution for 2M Clients & 2,000 Concurrent Connections

1. Executive Summary

This technical proposal outlines the software architecture, infrastructure design, agile team structure, licensing breakdown, and execution roadmap for the Limitless Naturals Mobile Application (iOS & Android) and Web Platform. Engineered for Eva Pharma, the solution integrates personalized AI health recommendations, chronic disease safety contraindications, a multi-channel admin notification center, real-time logistics management, and scalable cloud infrastructure for 2,000,000+ registered clients with 2,000 peak concurrent active sessions.

2. Proposed Infrastructure & Scale Specifications

Target Load: 2,000,000 Registered Clients | 2,000 Peak Active Concurrent Sessions

Metric / Layer	Proposed Infra Spec	Scaling Policy	Purpose / Rationale
DNS & CDN	Cloudflare Enterprise / AWS CloudFront	Edge Caching & WAF	DDoS protection & static acceleration
Load Balancer	AWS ALB Multi-AZ	Auto-SSL & Health Check	API traffic distribution
App Servers	AWS EKS 3x c6g.xlarge (4 vCPU, 8GB)	HPA: Auto-scale to 10 nodes	Node.js API & static web serving
Redis Cache	AWS ElastiCache 2x cache.m6g.large	Multi-AZ In-Memory Replication	Sub-5ms cart & session storage
Primary DB	AWS Aurora PostgreSQL (Primary + 1 Replica)	Auto-scaling Read Replica	ACID database for users & orders

3. Agile Scrum Team Structure (11 Specialists)

- **1x Product Owner (PO):** Defines business backlog & aligns feature prioritization with Eva Pharma.
- **1x Scrum Master (SM):** Facilitates 2-week Sprint ceremonies, daily standups, and removes impediments.
- **1x UI/UX Designer:** Crafts Figma design system, iOS/Android native mobile interfaces, and web layouts.
- **2x Mobile Developers:** Builds cross-platform Flutter iOS & Android apps with offline caching.
- **3x Backend Developers:** Node.js RESTful APIs, AI recommendation engine, payment & notification services.
- **1x DevOps Engineer:** CI/CD pipelines (GitHub Actions), Terraform IaC, Kubernetes, Grafana monitoring.
- **2x QC / QA Engineers:** Manual & automated API testing, security scanning, mobile device testing matrix.

4. Master Project Timeline (8 Months)

Project Phase	Duration	Schedule	Key Deliverables
Phase 1: Requirements	1 Month	Month 1	SRS, Figma designs, API contracts
Phase 2: Agile Core Dev	3 Months	Months 2–4	6 Sprints x 2 wks: Client, Admin & Distributor

Phase 3: SIT Testing	1 Month	Month 5	API integration, security & 2K load test
Phase 4: UAT Testing	2 Months	Months 6–7	Client UAT sign-off & feedback iterations
Phase 5: Go-Live & Support	1 Month	Month 8	App Store / Play Store launch & hypercare

5. Operational Cost Estimation & Third-Party Services

Service / Integration	Scope & Tooling	Estimated Cost
Cloud Hosting (AWS)	ALB, EKS, Aurora Postgres, ElastiCache, S3, CloudFront	~\$850 - \$1,400 / mo
Firebase (FCM)	Push Notifications (iOS & Android), Analytics	FREE (Unlimited)
Payment Gateways	Stripe / Fawry / PayMob / PayFort	2.2% - 2.75% + \$0.15/tx
Transactional Email	SendGrid / AWS SES (500,000 emails/mo)	~\$40 - \$80 / mo
SMS Gateway	Twilio / Local Telecom (~20,000 SMS/mo)	~\$200 - \$400 / mo
Developer Accounts	Apple Developer (\$99/yr) + Google Play (\$25 one-time)	~\$124 total

6. Milestone-Based Payment Structure

Milestone	Schedule	%	Trigger Condition / Deliverable
M1: Requirements	Month 1	15%	SRS & Figma designs approval
M2: Core Dev Phase A	Months 2–3	25%	Sprints 1–3: Client App, Auth & AI Engine
M3: Core Dev Phase B	Month 4	25%	Sprints 4–6: Admin Notifications & Distributor App
M4: SIT Testing	Month 5	15%	SIT & 2,000 concurrent load test sign-off
M5: UAT Testing	Months 6–7	10%	Client UAT sign-off & final tweaks
M6: Go-Live & Support	Month 8	10%	App Store / Play Store deployment & handover